Ahmed Refat Abd El-Moaty

Data Engineer

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https://www.linkedin.com/in/ahmedrefat1412 • https://github.com/AhmedRefat1412 • Portfolio

PROFESSIONAL SUMMARY

Data Engineer who builds batch and real-time pipelines on AWS, Azure, and Microsoft Fabric. Delivered Medallion lakehouses with Spark, Kafka, Apache Iceberg, and Delta Lake, with built-in data quality checks and ML inference. Certified: Fabric Data Engineer (DP-700), Azure Databricks Data Engineer (DP-750), AWS Solutions Architect.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy), SQL, PySpark, Bash
Processing & Streaming: Apache Spark, Spark Structured Streaming, Apache Kafka
Lakehouse & Warehousing: Delta Lake, Apache Iceberg, Snowflake, Amazon Redshift, Azure Synapse, TimescaleDB, Medallion Architecture, Star/Galaxy Schema
Cloud: AWS (S3, Glue, Redshift, Athena, EC2, IAM), Azure (ADLS, Data Factory, Event Hubs, Databricks), Microsoft Fabric (OneLake, Lakehouse, Warehouse, Eventstream, KQL, Real-Time Intelligence)
Orchestration & Quality: Apache Airflow, dbt, Azure Data Factory, Great Expectations
Databases: PostgreSQL, MongoDB, SQL Server
DevOps & BI: Docker, Git/GitHub, Linux, Power BI, Grafana

EXPERIENCE & TRAINING

Data Engineering Instructor | *Codix Academy, Remote* *Jul 2026 – Present*
- Designed a 48-session data engineering curriculum covering Python, SQL, Snowflake, Kafka, MongoDB, and AWS.
- Teach hands-on, project-based sessions on pipeline design, orchestration, and cloud data platforms.

Cloud Services Trainee | *National Telecommunication Institute (NTI), Alexandria* *Jan 2026 – Mar 2026*
- Provisioned and configured 6+ AWS services (EC2, IAM, Redshift, Glue, RDS, DynamoDB, S3) across 10+ lab environments.
- Applied Linux administration and networking fundamentals to support distributed data systems on AWS.

Data Engineer Trainee | *Microsoft – Digital Egypt Pioneer Initiative (DEPI), Cairo* *Nov 2025 – May 2026*
- Built Python and SQL ETL/ELT pipelines on Azure across 2+ projects, cutting data preparation time by 40% versus manual processing.
- Implemented Star and Galaxy Schema models across 3+ analytical domains.
- Completed a 6-month program covering ingestion, transformation, warehousing, and orchestration; delivered 2+ end-to-end cloud projects.

Mandatory National Service | *Egyptian Armed Forces* *Dec 2023 – Mar 2025*

TECHNICAL PROJECTS

EnergySentinel: Real-Time Energy Anomaly Detection GitHub ↗
Azure Event Hubs · Databricks · PySpark Structured Streaming · Delta Lake · Data Factory · Synapse Serverless SQL · LightGBM · Power BI
- Built a real-time Medallion pipeline (Bronze → Silver → Gold → Platinum) ingesting industrial sensor telemetry from Event Hubs.
- Ran real-time predictions with a pretrained LightGBM model inside Structured Streaming and stored results in the Platinum layer.
- Validated Silver data (nulls, duplicates, range and timestamp checks) and routed bad records to a Quarantine table; orchestrated with Data Factory and served a Power BI monitoring dashboard via Synapse Serverless SQL.

Predictive Telecom Streaming Lakehouse GitHub ↗
Apache Iceberg · AWS S3 · Kafka · PySpark Structured Streaming · Isolation Forest · Airflow · Athena · Grafana
- Architected a streaming lakehouse on Iceberg/S3 ingesting 4 concurrent API streams (system, network, radio, environmental) into dedicated Kafka topics.
- Joined all 4 streams in the Silver layer into unified Iceberg tables with ACID guarantees, schema evolution, and automated compaction.
- Applied Isolation Forest anomaly detection to each Gold micro-batch; a decoupled Kafka-consuming microservice sends Telegram alerts.
- Orchestrated 2 Airflow DAGs for Iceberg compaction and model retraining; built Grafana dashboards on Athena over the Gold layer.

Financial Data Platform: AWS Cloud Data Warehouse GitHub ↗
AWS Glue (PySpark) · S3 · Redshift · Airflow · dbt · Great Expectations · Power BI
- Built a Medallion ELT pipeline on AWS processing 2M+ financial transactions from raw S3 ingestion to fraud-analysis-ready Gold datasets.

- Enforced Silver-layer data quality with Great Expectations (schema, completeness, accuracy checks) to stop bad data reaching analytics.
- Modeled a Redshift star schema with Glue (PySpark) and dbt. orchestrated with Airflow DAGs and delivered a Power BI fraud dashboard.

Flight Delay Prediction Platform GitHub ↗

Kafka · PySpark Structured Streaming · TimescaleDB · Grafana · XGBoost · Airflow · Snowflake · Power BI

- Built a unified streaming and batch platform on 2 live APIs (OpenSky flights, Open-Meteo weather): Kafka → PySpark → TimescaleDB → Grafana.
- Embedded an XGBoost classifier in the streaming micro-batches for real-time delay prediction at 82% accuracy, with no separate serving layer.
- Built a 6-step Airflow batch DAG (Bronze → Silver → Gold → Snowflake) with a Galaxy Schema warehouse and 2 Power BI dashboards.

Global Supply Chain Control Tower

Fabric Data Factory · OneLake · Lakehouse · Warehouse · Delta Lake · Power BI

- Built an end-to-end analytics platform on Microsoft Fabric for a simulated enterprise (15 warehouses, 200 suppliers), unifying ERP, CRM, WMS, shipping, and weather sources into a Bronze → Silver → Gold Medallion architecture on OneLake.
- Cleaned and standardized cross-linked datasets, including supplier files with mismatched schemas, using incremental loading, CDC, and a quarantine path that isolates invalid records.
- Modeled a star schema in the Fabric Warehouse and delivered Power BI dashboards with row-level security for role-based access.

EDUCATION

B.Sc. Computer Science & Information Systems | *Kafr El-Sheikh University, Faculty of Computers and Information* *Graduated May 2023*

CERTIFICATIONS

Microsoft Certified: Fabric Analytics Engineer Associate (DP-600) *Nov 2026*
Microsoft Certified: Azure Databricks Data Engineer Associate (DP-750) *Sep 2026*
Microsoft Certified: Fabric Data Engineer Associate (DP-700) *Aug 2026*
AWS Certified Solutions Architect *Apr 2026*
Microsoft Data Engineer Certificate, Digital Egypt Pioneer Initiative (DEPI) *May 2026*

LANGUAGES

Arabic (Native) • English (Professional Working Proficiency, B2+)